

REPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie

MINISTERE DE L'ENSEIGNEMENT SUPERIEUR
COMMISSION NATIONALE D'ORGANISATION DES
EXAMENS ET CONCOURS

REPUBLIC OF CAMEROON
Peace-Work-Fatherland

MINISTRY OF HIGHER EDUCATION
NATIONAL COMMITTEE FOR THE
ORGANIZATION OF THE NATIONAL EXAMS AND
COMPETITIVE ENTRANCE EXAMINATIONS

National Exam of Higher National Diploma 2026 Session

Specialty/option: SWE – NWS – EDM
Paper : Discrete Mathematics
Duration : 4 hours

Credits : 4 / 4 / 10

GENERAL INSTRUCTIONS.

- **ANSWER ALL QUESTIONS**
- *You are reminded of the necessity of good English and orderly presentation of your material.*
- *Where calculations are required, clearly show your working and be chronological in your answer.*
- *You are authorized to use only non – programmable calculator*

SECTION – A: MULTIPLE CHOICE QUESTIONS (20 marks)

ANSWER THIS SECTION BY WRITING ONLY THE LETTER CORRESPONDING TO THE CORRECT ANSWER

1. If $y^2 = \cos^2 x$, then what will $-\frac{dy}{dx}$ be?
A: $-y \tan x$;
B: $y \tan x$;
C: $-y \cot x$;
D: $y \cot x$;
2. What is the value of $\tan\left(-\frac{3\pi}{4}\right)$?
A: -1 ;
B: $\sqrt{3}$;
C: 1 ;
D: $-\sqrt{3}$.
3. Find a positive value, c , for x , that satisfies the conclusion of Rolle's Theorem for Derivatives for the function $f(x) = x^2 - 2x - 15$ on the interval $[-3, 5]$.
A: 1 ;
B: $1/2$;
C: -3 ;
D: 5 .

4. For what value of x does the function $h(x) = \frac{(x-2)^2}{x^2-4}$ have a non-removable discontinuity?
- A: 1;
B: -1;
C: -2;
D: 2.
5. In certain grouped frequency distribution, two successive classes are 10.5 – 10.9 and 11.0 – 11.4. Which of the following correctly represents the lower class boundary of the class 11.0 – 11.4?
- A: 11.0.
B: 11.4.
C: 10.90.
D: 10.95.
6. If $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) then which of the following is TRUE about the Mean Value Theorem for Derivatives? There exists a point $c \in (a, b)$ such that,
- A: $f(c) = 0$;
B: $f'(c) = \frac{f(b)-f(a)}{b-a}$;
C: $f(c) = \frac{f(b)-f(a)}{b-a}$;
D: $f''(c) = 0$.
7. A vaccine has a success rate of 80 %. Five patients are administered the vaccine. Assuming the results are independent for all patients, what is the probability that at least one of them will be successfully cured?
- A: 0.9996;
B: 0.0004;
C: 0.4998;
D: 0.5002.
8. Suppose that the probability of an event X occurring is 0.70 and the probability of event Y occurring is 0.40. Also, suppose that the two events are independent. Then $P(A \cap B)$ is equal to:
- A: 1.10;
B: 0.28;
C: 0.55;
D: 0.83.
9. What name is given to a specific range of numbers within which a population mean should lie?
- A: Range;
B: Confidence coefficient;
C: Confidence interval;
D: Confidence level.
10. What is the sample mean of the following data? $\{X: x = 4, 5, 2, 6, 5, 2, 4\}$.
- A: 4.0;

- B: 5.0;
C: 2.0;
D: 6.0.
11. The Laplace Transform of $f(t) = 4 \cosh 3t$ is:
A: $3s/(s^2 - 9)$;
B: $4s/(s^2 - 9)$;
C: $3s/(s^2 + 9)$;
D: $s/(s^2 + 9)$.
12. The Inverse Laplace Transform of $\frac{s}{s^2 + 9}$ is:
A: $\sin 3t$;
B: $\sinh 3t$;
C: $\cosh 3t$;
D: $\cos 3t$.
13. The eigenvalues of the matrix $M = \begin{pmatrix} 1 & 4 \\ 3 & 2 \end{pmatrix}$ are:
A: $\{-3, -1\}$;
B: $\{2, 5\}$;
C: $\{1, 3\}$;
D: $\{-2, 5\}$.
14. Which of the following is the interval of convergence of the expansion of the function,
 $f(x) = \frac{x^2}{2-3x}$?
A: $\left(-\frac{2}{3}, \frac{2}{3}\right)$;
B: $\left[-\frac{2}{3}, \frac{2}{3}\right)$;
C: $\left\{-\frac{2}{3}, \frac{2}{3}\right\}$;
D: $\left[-\frac{2}{3}, \frac{2}{3}\right]$.
15. Suppose $r(x)$ is a function defined on the interval $[-4, 4]$ such that $\lim_{x \rightarrow -2} r(x) = -1$,
find $\lim_{x \rightarrow -2} [r(x)]^2$.
A: 2;
B: -2
C: 1;
D: -1.
16. If $g(x) = \sqrt{2-x}$ is expanded as a Maclaurin's series, then what is its range of convergence?
A: $-2 < x < 2$;
B: $-2 \leq x < 2$;
C: $-2 < x \leq 2$;
D: $-2 \leq x \leq 2$.
17. What is the equivalent of the trigonometric identity: $\operatorname{cosec}^2 x - \cot^2 x$?
A: $1 - 2 \tan^2 x$;

B: $\sin^2 x - \cos^2 x$;

C: $\cos^2 x - \sin^2 x$;

D: 1.

18. $\cos 2A$ is equal to

A. $\frac{1 - \tan^2 A}{1 + \tan^2 A}$

B. $\frac{1 + \tan^2 A}{1 - \tan^2 A}$

C. $\frac{1 - \tan^2 A}{2 \tan A}$

D. $\frac{1 + \tan^2 A}{2 \tan A}$

19. $\sin(A + B) - \sin(A - B) =$

A. $2 \sin A \cos B$

B. $-2 \cos A \sin B$

C. $2 \cos A \sin B$

D. $-2 \sin A \cos B$

20. If $y = \sqrt{x}$, the value of $\frac{dy}{dx}$ when $x = 9$ is

A. 0

B. $1/6$

C. $3/4$

D. 1

21. If $f(x) = \frac{1}{1-x}$, $f'(x) =$

A. $\frac{-1}{(1-x)^2}$

B. $\frac{-1}{x-1}$

C. $\frac{-1}{1-x^2}$

D. $\frac{1}{(1-x)^2}$

22. If $y = \sin 2x + \cos 2x$, $\frac{d^2 y}{dx^2}$ is

A. $-4 \cos 2x - 4 \sin 2x$

B. $4 \cos 2x + 4 \sin 2x$

C. $-4 \sin 2x + 4 \cos 2x$

D. $-\cos 2x - \sin 2x$

23. $\int (x^2 - \frac{1}{x^2} + \cos x) dx =$

A. $\frac{x^3}{3} + \frac{1}{x} - \sin x + k$

B. $\frac{x^3}{3} + \frac{1}{x} + \sin x + k$

C. $\frac{x^3}{3} - \frac{1}{x} + \sin x + k$

D. $2x + \frac{2}{x^3} - \sin x + k$

24. The first three terms of the series $\sum_{r=2}^{\infty} \frac{2^r}{r}$

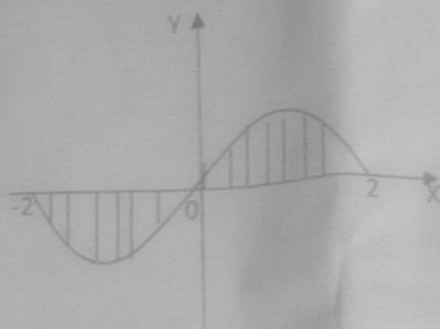
A. $0 + 2 + 4$

B. $2 + 8/3 + 4$

C. $2 + 8 + 32/3$

D. None of these

25. The shaded area in the diagram is given by



- A. $\int_{-2}^2 f(x) dx$
- B. $\int_{-2}^0 f(x) dx + \int_0^2 f(x) dx$
- C. $\int_{-2}^{-1} f(x) dx + \int_{-1}^2 f(x) dx$
- D. $\int_0^2 f(x) dx - \int_{-2}^0 f(x) dx$

26. A box contains 4 red balls, 5 blue balls and 3 green balls. Two balls are drawn simultaneously at random. The probability that both balls are of the same colour is:

- A. $\frac{4}{15}$
- B. $\frac{19}{66}$
- C. $\frac{8}{33}$
- D. $\frac{1}{6}$

27. The domain of the function $f(x) = \ln(2 - x)$ is:

- A. $(-\infty, 2)$
- B. $(2, \infty)$
- C. $(-\infty, \infty)$
- D. $[2, \infty)$

28. A biased coin has probability $P(\text{head}) = \frac{2}{3}$. The coin is tossed once. Let X be the random variable defined by

$$X = \begin{cases} 1, & \text{if head appears} \\ 0, & \text{if tail appears} \end{cases}$$

The variance of X is :

- A. $\frac{1}{9}$
- B. $\frac{2}{9}$
- C. $\frac{4}{9}$
- D. $\frac{5}{9}$

29. Which of the following values of the correlation coefficient r indicates the strongest positive linear relationship between two variables?

- A. $r = -9.92$
- B. $r = 0.10$

- C. $r = 0.85$
 D. $r = 1.50$
30. Evaluate: $\sum_{k=1}^8 2k$
 A. 72
 B. 48
 C. 56
 D. 64
31. Let $a_n = \frac{(-1)^n}{n+1}$, where $n \in \mathbb{N}$. Then $\lim_{n \rightarrow \infty} a_n$ is:
 A. -1
 B. 0
 C. 1
 D. Does not exist
32. Let $f(x) = x^4 - x^2$. Then the function f is:
 A. Even
 B. Odd
 C. Periodic
 D. Neither even nor odd
33. Let X be a continuous uniform random variable on the interval $[a, b]$, with $a < b$. The mean of X is:
 A. $\frac{a+b}{2}$
 B. $\frac{b-a}{2}$
 C. $\frac{(b-a)^2}{12}$
 D. $\frac{b^2-a^2}{2}$
34. Evaluate: $\int_0^2 (x+1) dx$
 A. 2
 B. 3
 C. 4
 D. 6
35. The value of the constant k such that $\int_0^2 kx dx = 8$ is:
 A. 1
 B. 2
 C. 3
 D. 4
36. Given the sequence (U_n) defined by $U_n = \frac{1}{n}$ the sequence is:
 A. Divergent
 B. Convergent
 C. Increasing
 D. constant
37. evaluate: $\int e^x \cos x dx$
 A. $\frac{e^x}{2}(\sin x + \cos x) + c$
 B. $e^x(\sin x - \cos x) + c$
 C. $e^x \cos x + c$

D. $\sin x + c$

38. Evaluate: $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$

A. 0

B. $\frac{1}{2}$

C. 1

D. Does not exist

39. Evaluate: $\int_0^{\frac{\pi}{2}} \cos x \, dx$

A. 0

B. 1

C. 2

D. $\frac{\pi}{2}$

40. If y is a solution of $\frac{dy}{dx} = 3x^2$, then y is:

A. $x^3 + c$

B. $x^2 + c$

C. $3x^3 + c$

D. $3x^2 + c$

SECTION B: STRUCTURAL (80 MARKS)

I. Analysis (30 Marks)

- 1) State Rolle's Theorem. Find the value of x_0 prescribe in Rolle's Theorem for $f(x) = x^3 - 12x$ on the interval $0 \leq x \leq 2\sqrt{3}$ [5 marks]
- 2) Evaluate the double integral below. [5 marks]

$$\int_{-1}^2 \int_{2x^2-2}^{x^2+x} x \, dy \, dx$$

- 3) Find the discontinuities of the following functions. Determine whether they have removable or jump discontinuities.

a) $f(x) = \frac{x-1}{(x+3)(x-2)}$

b) $f(x) = \frac{(x+2)(x-1)}{(x-3)^2}$

c) $f(x) = \frac{x^3-27}{x^2-9}$

d) $f(x) = \frac{4-x^2}{3-\sqrt{x^2+5}}$ (4 x 2 = 8 marks)

- 4) Solve the differential equation $y'' - y' + 2y = 0$, given that $y(0) = 1, y'(0) = 0$. [6 marks]

- 5) Given $\mathbf{B} = xy^2\mathbf{i} + 2x^2yz\mathbf{j} - 3yz^2\mathbf{k}$, that
Find

a) Div B (divergence of B)

b) Curl B

[2 x 3 = 6 marks]

II. Statistics (30 Marks)

The weight y in kg , of a set of students was measured with respect to their heights, x in cm . The results were:

Height (x/cm)	165	176	159	167	174	171	169	168	169	172
Weight (y/kg)	72	90	70	75	86	84	80	81	82	83

- Plot a graph of y against x showing these points (the graph is called a scatter diagram)
- Determine the equation of the regression line of y on x .
- Determine the equation of the regression line of x on y .
- Plot the regression lines in (b) and (c) on the same graph as the scatter diagram in (a).
- Calculate the product moment correlation coefficient for the distribution.
- Calculate the Spearman's correlation coefficient of rank.
- Calculate the Kendall correlation coefficient of ranks

[4, 6, 2, 4, 6, 4, 4 marks]

III. Probability (20 Marks)

- If A, B, and C are mutually exclusive events with $P(A) = 0.2$, $P(B) = 0.3$, and $P(C) = 0.4$, determine the following probabilities :
 - $P(A \cup B \cup C)$,
 - $P(A \cap B \cap C)$,
 - $P(A \cap B)$,
 - $P[(A \cup B) \cap C]$,
 - $P(A' \cap B' \cap C')$

[10 marks]
- 3.2 A part selected for testing is equally likely to have been produced on any one of six cutting tools.
 - What is the sample space?
 - What is the probability that the part is from tool 1?
 - What is the probability that the part is from tool 3 or tool 5?
 - What is the probability that the part is not from tool 4?

[2, 2, 3, 3 marks]